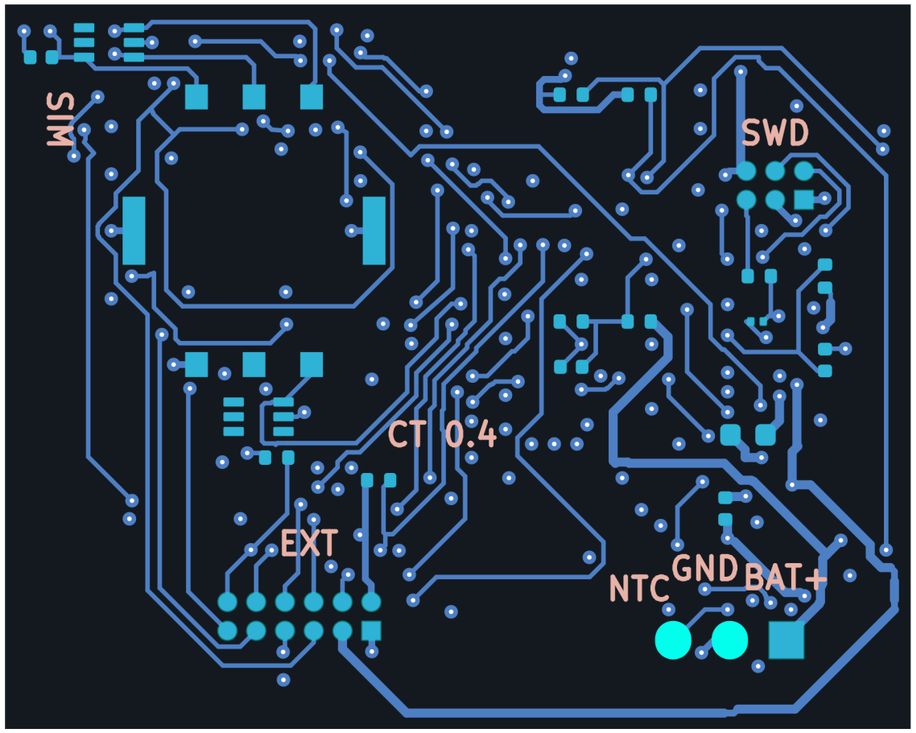
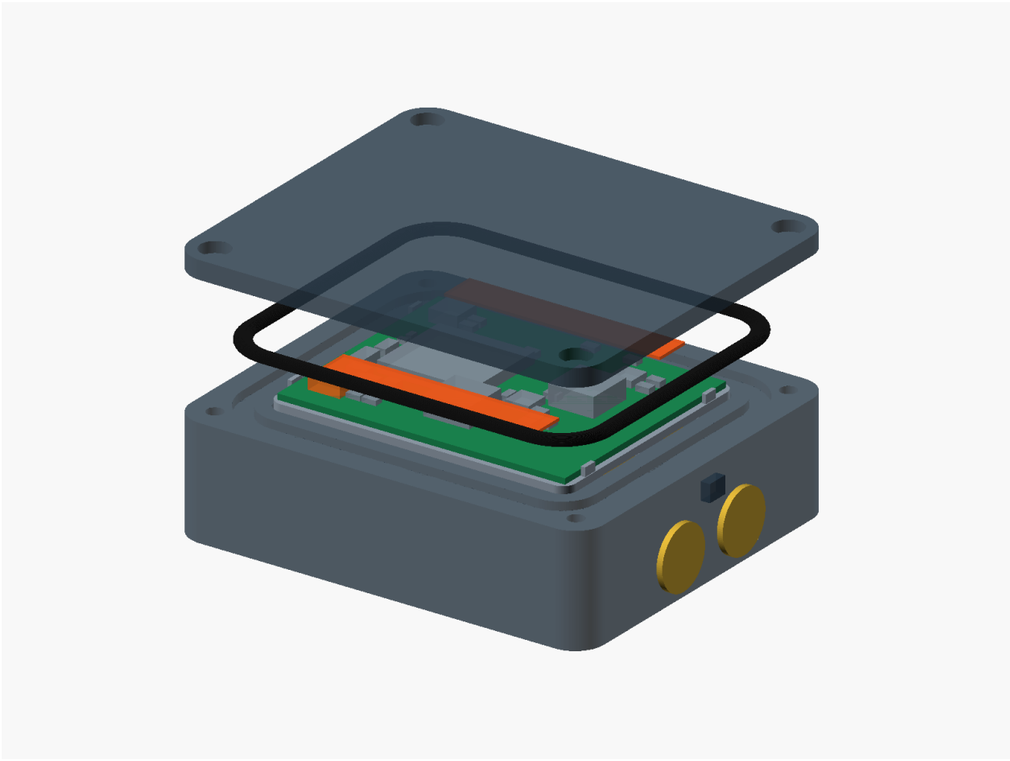
# COLLAR TRACKER / DESIGN REVIEW

r04i PCB verified • swim-r02 populated-fit prototype • final part selections and physical tests remain open

## 40 × 32 mm / FOUR-LAYER PCB




## SEALED HOUSING / EXPLODED MODEL



## PCB CHECKS COMPLETE

Native DRC / ERC / connectivity / parity: zero findings

Two real JLC records: 0 Danger / 0 Warning each

Identical ZIP, unchanged rules, 29 analyzed rows each

82 local GND pads; two continuous inner GND layers

93 / 93 signal-via return screens; SIM lane clear

SMT assembly, power and RF qualification remain open

## MECHANICAL PROTOTYPE

Rigid body: 57.4 × 49.4 × 17.6; 58.7 including the key

With collar boot: 60.4 × 52.4 × 22.4

Continuous lid O-ring; two independent contact seals

Target: 2 m fresh water / 60 min — provisional conditions

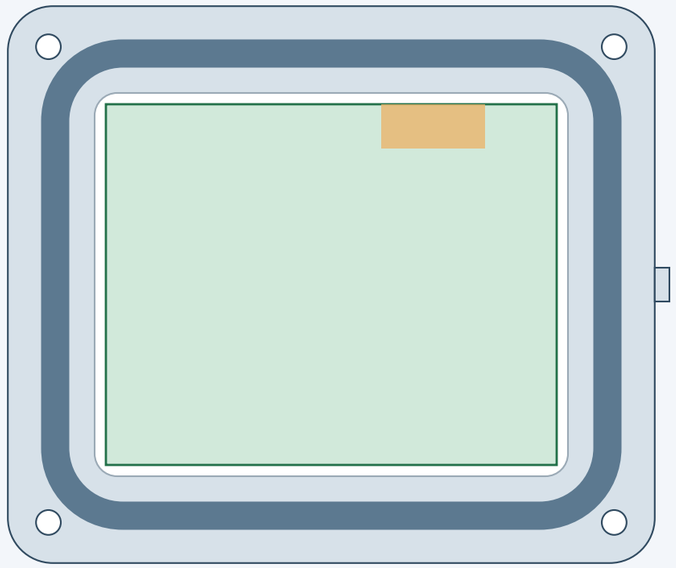
49 component heights checked; pack and antennas still open

Carrier reliefs and M3 contact retention; no physical IP test

# SWIM-r02 / DIMENSIONED PROTOTYPE

Nominal CAD plus controlled seal dimensions • CNC/process and material approval required before manufacture

## TOP VIEW / LID REMOVED



57.4 body width; 58.7 with key

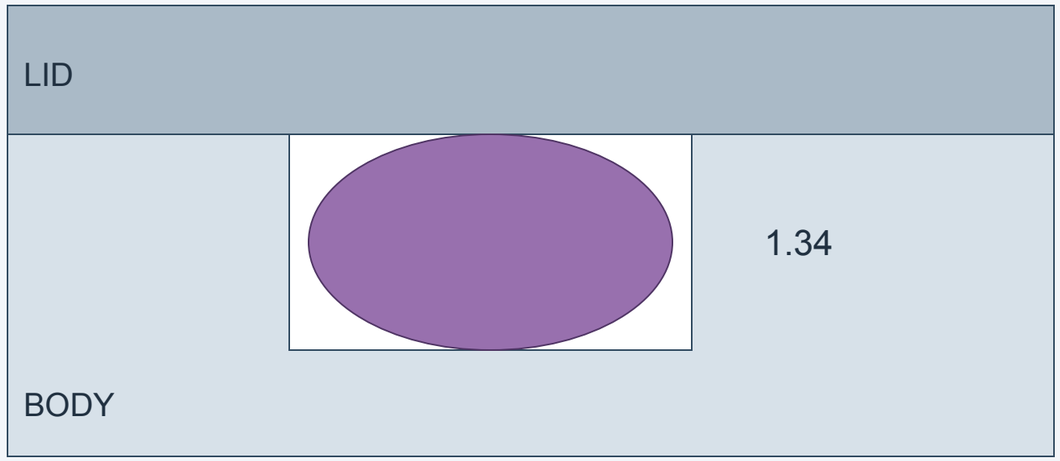
49.4 body depth • R4 outer corners

42 × 34 cavity, R2 • PCB 40 × 32

4 × Ø2.2 at (±25.1, ±21.1)

Orange: SIM lane; lift tray for card service

## MAIN LID SEAL / SECTION




2.50 ±0.05

Depth 1.34 ±0.02; local flatness budget 0.01

Centerline 49 × 41, R6; bottom corner R0.2

O-ring 50.52 ID × 1.78 section, EPDM 70A

Declared squeeze: 17.7–28.8%; fill: 62.6–83.2%

Seal face Ra ≤0.8 µm; no burrs or crossing tool marks

## SEALED CHARGING TARGET / CUSTOM TURNED PART

Head Ø8.5 × 0.8 | Shaft Ø7.28 × 7.70 | Rear M3 × 0.5 thread, 3.50 long

Radial gland root Ø4.64 ±0.01; width 2.80 ±0.05; center 3.85 behind head

Housing bore Ø7.34 ±0.02; seal ID 4.50 ±0.10 × section 1.80 ±0.08

Centers at Y=±6, Z=5.5 on +X wall. +Y is +5 V; −Y is GND.

Contact coating, edge finishing, retention, dock parts and electrical/immersion durability require prototype qualification.